

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	5

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Code of Conduct:

*I **Kadhirezhilan. D [192511228]** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

1. Which of the following is true about 1NF?

Answer: (c) All attributes must be atomic

Reason: First Normal Form requires every attribute to contain **atomic (indivisible) values**. It does not allow repeating groups or multivalued attributes.

2. A relation is in 2NF if it is in 1NF and:

Answer: (b) Has no partial dependencies on primary key

Reason: 2NF eliminates **partial functional dependencies**, meaning every non-prime attribute must depend on the **whole candidate/primary key**, not just a part of a composite key.

3. Which normal form deals with multi-valued dependencies?

Answer: (d) 4NF

Reason: Fourth Normal Form (4NF) specifically addresses **non-trivial multivalued dependencies (MVDs)** and reduces redundancy caused by independent multivalued facts.

4. BCNF is a stricter version of:

Answer: (c) 3NF

Reason: Boyce-Codd Normal Form (BCNF) is stronger than 3NF. In BCNF, for every non-trivial functional dependency:

$$X \rightarrow Y$$

X must be a superkey.

5. Functional dependency $X \rightarrow Y$ means:

Answer: (b) X determines Y

Reason: $X \rightarrow Y$ means that if two tuples have the same value of **X**, they must have the same value of **Y**.

In simple terms:

X determines Y.

Example:

StudentID $\rightarrow$ StudentName

A particular StudentID determines exactly one StudentName.

6. Which anomaly is NOT associated with unnormalized relations?

Answer: (d) Retrieval anomaly

Reason: The classical anomalies caused by poor/unnormalized database design are:

- **Insertion anomaly**
- **Update anomaly**
- **Deletion anomaly**

"Retrieval anomaly" is not one of the standard three normalization anomalies.

7. A superkey is:

Answer: (b) Any set of attributes that uniquely identifies tuples

Reason: A superkey is any attribute set that uniquely identifies each tuple.

For example, if StudentID uniquely identifies a student:

{StudentID} is a superkey.

{StudentID, StudentName} can also be a superkey, although it is not minimal.

A candidate key is different because it must be a **minimal superkey**.

Therefore:

Superkey = uniquely identifies tuples

Candidate Key = minimal superkey

8. If $R(A,B,C)$ has FD $A \rightarrow B$ and $B \rightarrow C$, then C is:

Answer: (c) Transitively dependent on A

Reason:

Given:

$A \rightarrow B$

and

$B \rightarrow C$

by transitivity:

$A \rightarrow C$

Therefore, C is **transitively dependent on A through B**.

The dependency chain is:

$A \rightarrow B \rightarrow C$

9. 5NF is also known as:

Answer: (c) Project-Join Normal Form

Reason: 5NF is called Project-Join Normal Form (PJ/NF).

It deals primarily with **join dependencies** and ensures that a relation cannot be further decomposed into smaller relations without losing information or introducing spurious tuples.

10. Lossless-join decomposition ensures:

Answer: (a) No data is lost during decomposition

Reason: A decomposition is **lossless** when joining the decomposed relations reconstructs the original relation without losing information or generating incorrect/spurious tuples.

Conceptually:

Original Relation → Decompose → Join Back → Original Relation

Therefore, the correct answer is **(a)**.

RUBRICS FOR CALCULATION:

MCQ					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions